VEHICULAR ACCESS ~~TRAFFIC~~, TRANSPORT IMPACT, PARKING AND BICYCLES

SUTHERLAND SHIRE

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Vehicular Access Traffic Parking and Bicycles

This chapter supplements the vehicular access, traffic, parking and bicycle controls provided for specific types of developments throughout the DCP. It provides detailed controls which apply across most forms of development and establishes what Council will consider acceptable design solutions in regard to these issues. It also sets controls for the subdivision of land.

This chapter must be read in conjunction with the development specific controls contained in the relevant chapter of this DCP. In addition the following environmental planning instruments, Council documents, Australian Standards and other guidelines apply as relevant:

Environmental planning instruments:
State Environmental Planning Policy (Transport and Infrastructure) ~~2007~~2021
Sutherland Shire Local Environmental Plan 2015

Council documents:
Sutherland Shire Public Domain Design Manual

Australian Standards and other Guidelines:
AS2890.1: Parking Facilities – Off-street Car Parking
AS2890.2: Parking Facilities – Off-street Commercial Vehicle Facilities
AS2890.3: Bicycle parking facilities
AS2890.6: Off street car parking for people with disabilities
AS4299 - 1995: Adaptable Housing
TfNSW Guide to Transport Impact Assessment ~~RTA Guide to Traffic Generating Developments~~
Austroad Guide to Traffic Engineering Practice – Part 14 – Bicycles

Parking

Accommodating vehicles can have a significant impact on the design of new development. The location, layout and design of the parking will influence the layout and design of buildings and landscaping.

Development must provide sufficient on-site parking to meet the demand it generates. Parking is required for different types of vehicles according to the proposed use. Vehicles include passenger vehicles, motor bikes, light vehicles and heavy vehicles and pushbikes.

The provision of on and off street parking must not compromise vehicles, pedestrian or cyclist safety.

1. Number of Parking Spaces

1.1 Objectives

1. Ensure all land uses and/or combination of activities provide sufficient parking on site to satisfy the demand for parking by different vehicle types generated by the development, and minimise reliance on on-street parking.

2. ~~Minimise reliance on street parking.~~To provide for an efficient and safe transport network that caters for all users.

3. To encourage mode shift where there is good access to shops, services and public and shared transport options.

4. To encourage the use of alternative means of transport and the provision of end-of-trip facilities.

5. To manage parking supply at both trip origins and destinations to reduce private vehicle use.

~~2.~~6. To provide convenient and safe on-site parking and drop-off area where commensurate and suitable for the land uses.

1.2 Controls

1. Car parking ~~shall~~ must be provided in accordance with the car parking requirements set out in Table 1.

2. Where the car parking requirement is expressed as a minimum number of spaces the development ~~shall~~ must not provide less spaces than that minimum.

3. When the calculations for the numbers of parking spaces results in a part or fraction of a parking space of 0.5 or greater for the whole development, then the actual number ~~shall~~ must be rounded up to the greater whole number. For ~~example~~example, 1.5 spaces ~~shall~~ must be rounded up to 2 spaces for the whole development.

4. Where a development proposal contains two or more land uses the number of parking spaces ~~requirement shall~~ must be the sum of parking required for the individual land uses. However, where the proposed land uses comprise different peak parking demands, the total requirement

may be reduced such that the peak demand is met at any one time, where supported by a study by a suitably qualified traffic engineer.

5. In the following circumstances, a Transport Impact Assessment must be provided:

- Where car parking requirements for a proposed development are not listed in the DCP, or
- ~~w~~Where the development proposal raises unique traffic and parking issues, or
- ~~5.~~ ~~w~~Where development is identified as Traffic Generating Development, as defined by the State *Environmental Planning Policy (Transport and Infrastructure) 2021.* ~~then a Traffic Report~~Transport Impact Assessment ~~shall be completed.~~

~~6. A Traffic Report~~Transport Impact Assessment ~~is also required for a Traffic Generating Development as defined by State *Environmental Planning Policy (Transport and Infrastructure) 2007*~~2021~~.~~

~~7.~~6. Motor cycle parking ~~shall~~ must be provided for all non residential development at a rate of 1 motorcycle space per 25 car spaces or part thereof. For ~~example~~example, where 26 car parking spaces are required then 2 motorbike parking spaces are to be provided. Motor cycle parking spaces ~~shall~~ must comply with the relevant standards.

Table 1 – Car Parking Numbers

Use	Rate
Residential Accommodation	
Dwelling Houses	Minimum: 2 spaces per dwelling Maximum: 43 spaces per dwelling * Only 2 spaces are to be evident when viewed from the street **
Secondary Dwellings	No requirement
Dual Occupancies	Minimum 1 space per dwelling Maximum 3 spaces per dwelling.* Only 2 spaces are to be evident (per development) when viewed from the street
Multi Dwelling Housing	~~Zones R2, R3, R4, B1, & B2~~ E1 ~~Minimum 1 space per 1 bed; plus,~~ ~~1.5 spaces per 2 bed; plus,~~ ~~2 spaces per 3 bed; plus,~~ ~~1 visitor space per 4 dwellings~~ ~~Maximum: up to 3 spaces per dwelling*~~ **Zone R2, R3, R4~~, E1~~** Miranda and Sutherland within 800m of rail station 0.8 space per 1 bed 1.0 space per 2 bed 1.5 space per 3 bed 1 visitor space per 7 units Cronulla, Kirrawee, Caringbah within 800m of rail station 1.0 space per 1 bed 1.0 space per 2 bed 1.5 space per 3 bed 1 visitor space per 5 units All other areas 1.0 space per 1 bed 1.3 space per 2 bed 1.8 space per 3 bed 1 visitor space per 5 units **Zone E1, E2, MU1** Minimum 1 space per unit – No visitor parking Maximum: up to 3 spaces per dwelling for all zones*
Residential Flat Building/Shop Top Housing	~~Zones R4 & R3~~ ~~Minimum: 1 space per 1 bed, 1.5 spaces per 2 bed, 2 spaces per 3 bed, 1 visitor space per 4 units~~ ~~Zone B1, B2, B3 & B4~~ E1, E2 and MU1

	~~Minimum 1 space per unit – No visitor parking~~ ~~Maximum: up to 3 spaces per dwelling*~~ **a) Zone R2, R3, R4~~, E1~~** i. Miranda and Sutherland within 800m of rail station 1. ~~0.8~~1.0 space per 1 bed 2. ~~1.0~~2.0 space per 2 bed 3. ~~1.5~~2.5 space per 3 bed 4. 1 visitor space per ~~7~~ 4 units ii. Cronulla, Kirrawee, Caringbah within 800m of rail station 1. ~~1.0~~1.0~~-~~space per 1 bed 2. 2.0~~1.0~~ space per 2 bed 3. 2~~1~~.5 space per 3 bed 4. 1 visitor space per 4~~5~~ units iii. All other areas 1. 1.0~~1.0 ~~space per 1 bed 2. 2.0~~1.3~~ space per 2 bed 3. 2.5~~1.8~~ space per 3 bed 4. 1 visitor space per 4~~5~~ units **b) Zone E1, E2, MU1** 1. 1 space per 1 bed 2. 2 space per 2 bed 3. 2.5 space per 3 bed 4. 1 visitor space per 4 units ~~Minimum 1 space per unit – No visitor parking~~ ~~Maximum: up to 3 spaces per dwelling for all zones*~~
Seniors Housing	~~Car parking rates consistent with~~As per State Environmental Planning Policy (Housing) 2021 (Part 5 Housing for Seniors ~~or~~ and People with a Disability) ~~2004 where the SEPP does not otherwise apply~~
Boarding House (not subject to ARH SEPP)	Zone ~~B3~~ E2 1 space for every 2 bedrooms; plus, 1 space for any residential manager; plus 1 space for every 2 employees working at any one time. All other zones: 1 space for every 2 bedrooms; plus, 1 space for any residential manager; plus, 1 space for every 2 employees working at any one time.
Tourist and Visitor Accommodation – minimum rates	
Bed & Breakfast accommodation	1 space per guest room Note –This parking requirement is in addition to the parking required for the dwelling house.

Hotel or Motel accommodation	1 space per 4 rooms; plus, 1 space per 2 employees
Serviced Apartments	1 space per 2 units; plus, 1 space per 2 employees
Backpackers' Accommodation	~~Traffic Report~~Transport Impact Assessment
Commercial Premises – minimum rates	
Office and Business Premises	~~Zones B1, B2, B3 & B4~~ E1, E2 and MU1~~: 1 space per 30m^2 GFA~~ ~~Zones B5 & B6~~ E3~~: 1 space per 45m^2 GFA~~ Zones E1, E2, MU1, and E3 Miranda and Sutherland within 800m of rail station: 1 space per 60m2 GFA All other Zones E1, E2, MU1 and E3: 1 space per 40m2 GFA
Retail Premises	~~Zones B1, B2, B3 & B4~~ E1, E2 and MU1~~: 1 space per 30m^2 GFA~~ ~~Zones B5 & B6~~ E3~~: 1 space per 45m^2 GFA~~ Zones E1, E2, MU1, and E3 Miranda and Sutherland within 800m of rail station: 1 space per 60m2 GFA All other Zones E1, E2, MU1 and E3: 1 space per 40m2 GFA
Land Use Terms Outside Commercial Premises Group Term – minimum rates	
Shopping centres and supermarkets	Refer to Guide to Transport Impact Assessment, including a reference to a study of a similar-sized and contextually located site.
Industrial retail outlets	1 space per ~~-~~45m^2
Restricted premises	1 space per 30m^2 GFA
Sex services premises	1 space / 2 employees (non-sex workers); plus, 1 space per sex worker on the premises at any one time, or per work room, whichever is greater.
Service stations:	6 spaces per service bay. 1 space per 30m^2 of any convenience store within the service station. 1 space per 8m^2 of gross floor area for any food shop within the service station. Where no service / work bays, 1 space per 2 employees.
Pubs Registered Clubs	Refer to ~~RTA Guidelines for Traffic Generating Development~~ Transport for NSW Guide to Transport Impact Assessment
Industrial Premises – minimum rates	
	1 space per 100m², with a minimum of 2 spaces for each industrial unit. Any ancillary office component to an industrial development ~~shall~~ must provide 1 space per 30m² of

	gross floor area.
Land Use Terms Outside Industry Group Term – minimum rates	
Boat building and repair facilities	~~Traffic Study~~Transport Impact Assessment
Vehicle body repair workshops Vehicle repair stations	4 spaces/service bay.
Heavy Industrial Storage Establishment Hazardous storage establishment, Liquid fuel depots Offensive storage establishments	1 space per 100m², with a minimum of 2 spaces for each industrial unit. Any ancillary office component to an industrial development ~~shall~~ must provide 1 space per 30m² of gross floor area.
Storage Premises – minimum rates	
Self storage units; Transport depots Wharf or boating facilities	~~Traffic study.~~Transport Impact Assessment
Depots Warehouse or distribution centres	1 space per 300m², with a minimum of 2 spaces for each industrial unit. Any ancillary office component to an industrial development ~~shall~~ must provide 1 space per 30m² of gross floor area. 1 space per 30m² for any ancillary shop/retail space.
Educational Establishment	
Schools	~~Traffic Study~~Transport Impact Assessment
Health Services Facilities – minimum rates	
Medical Centres	Zones ~~B1, B2, B3,B4~~ E1, E2 and MU1 1 space per 30m² Caringbah Medical Precinct (R4) 1 space per 35m²
Health consulting rooms	3 spaces per consultation room or surgery room
Other Uses – minimum rates	
Childcare Centres	1 space per ~~4~~ 5 children in attendance, with at ~~least~~maximum 50% parking allocated to staff. ~~Provision for flexibility if :~~ ~~- centre is near a public reserve~~ ~~- centre is located on a corner block~~ ~~- centre provides a safe drop off zone on the street~~ ~~- the centres has a street frontage greater than 15m, or~~ ~~- if the centre operates as a long day care centre.~~
Amusement Centres, Entertainment facilities, Function centres Highway service centres, Registered clubs Veterinary hospitals Mortuaries Places of public worship Respite day care centres Jetties, Recreation facilities (indoor)	~~Traffic Study~~Transport Impact Assessment

Recreation facilities (outdoor)	
Marinas	1 space for every 3 berths (wet and dry); plus, 1 space for every 2 employees on duty at any one time.
Caravan Park Ecotourist Facility	1 space per caravan site or ecotourist unit. 1 visitor space per 20 caravan site or ecotourist unit. 1 space for site manager.

*Up to a maximum of four (4) car parking spaces provided outside of a basement are to be excluded from the calculation of Gross Floor Area, provided all existing objectives and controls for car parking and site design are metWhere more than the minimum parking spaces are proposed per dwelling, the additional space/s will only be considered to meet Council's requirement for parking, andparking and be excluded from the calculations of gross floor area, if it is provided within a basement and meets the objectives and controls for basement specified in the relevant SSDCP2015 chapter.

**All basement car parking spaces are to be excluded from the calculation of Gross Floor Area, regardless of the number of spaces, provided all existing objectives and controls for basements are met.

Maximum parking rates in a basement meet Councils requirements for parking, and as such are not included in the calculations of gross floor area.

2. Active Transport and Car Share

Active transport includes walking, cycling and public transport. Increased housing and urban density around centres can worsen traffic congestion, parking availability and negatively impact pedestrian safety. Encouraging active transport options over vehicle use for trips within well-connected centres will improve amenity and benefit the environment.

Green Travel Plans include measures aimed at promoting sustainable travel methods for occupants and visitors. Car share schemes reduce the demand for private vehicle ownership by providing a self-service car rental scheme. Car share is an increasingly popular transport option, providing convenience in areas where there is a good level of walking, cycling and public transport options.

2.1 Objectives

1. Promote alternative options for active transport to manage vehicle use around town centres

2. Sustainably manage traffic demand for large scale developments

3. Provide opportunities for car share schemes within developments to reduce the demand on private vehicle ownership.

4. Car share is encouraged in residential flat buildings, shop top housing and large commercial developments, where public transport is in close proximity.

2.2 Controls

1. A Green Travel Plan must be prepared for all ~~commercial~~non-residential and shop top housing developments:
   a. Over 2000m^2 in gross floor area ~~for commercial land uses~~
   b. For new or significant alterations to educational facilities
   c. Commercial developments with 50 or more occupants (~~[~~staff and visitors~~?]~~)

   ~~—~~Development generating significant traffic, including Traffic Generating Development, as defined by State *Environmental Planning Policy (Transport and Infrastructure) 2021.* ~~As deemed necessary by Council.~~

2. A Green Travel Plan must include at a minimum:
   - Targets - objectives and mode share targets (including timeframes for implementation)
   - Travel data - resident, visitor and workforce data analysis to inform proposed strategies
   - Measures– strategies, tools or actions to achieve the targets (to include communication methods for site occupants)
   - Monitoring – implementation and monitoring pre and post development.

~~Car share is encouraged in residential flat buildings, shop top housing, and large commercial developments where public transport is high.~~

3. Council may consider car share spaces ~~concessions~~ in lieu of some resident or commercial parking spaces, at a rate of 1 per 3 spaces, given they meet all the following criteria:

a. Residential Flat Building/Shop Top Housing, or Commercial developments over 2000m^2 GFA
b. Within 800m of a railway station at selected centres (Miranda, Sutherland, Kirrawee, Caringbah, Cronulla)
c. Cannot replace more than 25% of the minimum off-street parking requirement, with a maximum of 4 car share spaces per development

4. Car share spaces must be:
   a. Publicly accessible 24/7 in a well-lit location with safe and convenient pedestrian access
   b. Located in a mobile phone signal area to ensure connectivity to the car share scheme network
   c. Located adjacent to electrical infrastructure to allow for EV charging points
   d. Clearly designated as a car share space by signage and line markings.'

~~2.~~5. Evidence of an arrangement with an appropriate car share operator ~~is to~~must be provided along with a management plan detailing operation, security, and access.

3 Electric Vehicle Charging Infrastructure

More electric vehicles on the road results in lower carbon emissions and improved localised air quality. As the uptake of electric vehicles grows in the Sutherland Shire, infrastructure for vehicle charging should be considered at the design phase for new developments. This will assist with the transition to electric vehicles as we move towards a low carbon future.

3.1 Objectives

1. Provide the electrical infrastructure required to facilitate increased future usage of electric vehicle charging equipment in residential and non-residential development.

3.2 Controls

1. Installation of EV charging equipment with a minimum rated power output of 7kW (32A) is encouraged for new dwelling houses, semi-detached dwellings or dual occupancies.

2. All off-street car parking spaces in new residential and non-residential developments are to have electrical infrastructure to accommodate EV charging. This is to include:
   - Adequate electrical capacity and infrastructure for the installation of EV charging equipment in accordance with NCC requirements.
   - Electric circuitry for EV charging equipment is required to be a minimum rated power output of 7kW (32A single phase).

3. Where a development has non-private parking spaces, a number of non-private parking spaces shall have electric vehicle charging equipment installed.
   a) The minimum number of non-private parking spaces for which electric vehicle charging equipment shall be installed will be the lesser of:
      - 10% of the total number of all parking spaces in the development, or
      - the total number of non-private parking spaces.
      - Numbers are rounded up, if 0.5 or more to the nearest whole number.

4. In accordance with EV Distribution Board requirements of NCC, identify the wiring enclosure system to allow each car space to install an electric vehicle charger point. This system ~~should~~must allow future installation of cabling to electric vehicle charger points and allow internet access using one of the following:
   a) Cable tray of a size designed and calculated to be sufficient shall be installed from the EV Distribution Board to within 5m of all parking bays allocated for in the EV Distribution Board on that storey.
   b) An alternative to a cable tray may be provided, if it demonstrates the same or better than the requirements of provisions 4a above.

5. Electric vehicle charging equipment installed ~~shall~~must include a billing mechanism for residents and visitors using the electric vehicle charging equipment. The developer defines for the owners corporation the means by which sub-billing or cost allocation may be achieved by the incoming owners corporation when they take over operation of the site. These means shall not constitute vendor lock-in – i.e. there must be a clear pathway for the owners corporation to make alternative billing arrangements without a punitive exit clause from a service provider.

6. The following information should accompany a development application:
   a) Car parking spaces with EV charging equipment identified on architectural plans,
   b) EV charging equipment and infrastructure must be identified on an electrical plan, or equivalent; and
   c) a statement prepared by a suitably qualified person must be provided which specifies that the proposal will be able to meet the electric vehicle requirements of this Development Control Plan and the National Construction Code.

7. Electric vehicle charging equipment shall be compliant with Australian Standards, marked with Regulatory Compliance Mark (RCM) markings, and shall be equipped with OCPP 1.6J (or higher) communications capability.

8. All works will be undertaken by appropriately licenced persons and in keeping with relevant standards and regulations including Section J9D4 'Facilities for electric vehicle charging equipment' in NCC 2022 Volume One - Building Code of Australia that contains details for provision and design of distribution boards to support electric vehicle charging in car parks.

Note:
Private car spaces are defined as any car parking space that is on the title of a dwelling.
Non-private car spaces exclude the private car spaces and include all other car spaces (for example provided for visitors, contracted car share companies and service vehicles).

4. Design of Car Parking Areas and Basements

4.1 Objectives

1. Provide safe and easy access to and from the site for all vehicles and active transport modes.

2. Minimise pedestrian and vehicle conflicts.

3. Ensure access where pedestrians and/ or drivers may have a disability or mobility constraint (e.g., accompanied by young children).

4. Provide adequate turning areas for manoeuvring into and out of car parking spaces and/or garages.

5. Ensure minimal disruption of vehicles on public roads.

6. Ensure that all elements of development visible from the street, waterways and public domain make a positive contribution to the foreshore, streetscape and natural features of the area.

7. Ensure development is compatible with the scale, character and landscape setting of the adjoining streetscape, natural setting and scenic quality and that the environment's natural qualities dominate.

8. Ensure basements are sympathetic to the natural landform including natural rock outcrops and topographical features of the site and minimise the extent of excavation.

9. Basements must not compromise the safety of the on-street and off-street environment for vehicles, pedestrians or cyclist.

4.2 Controls

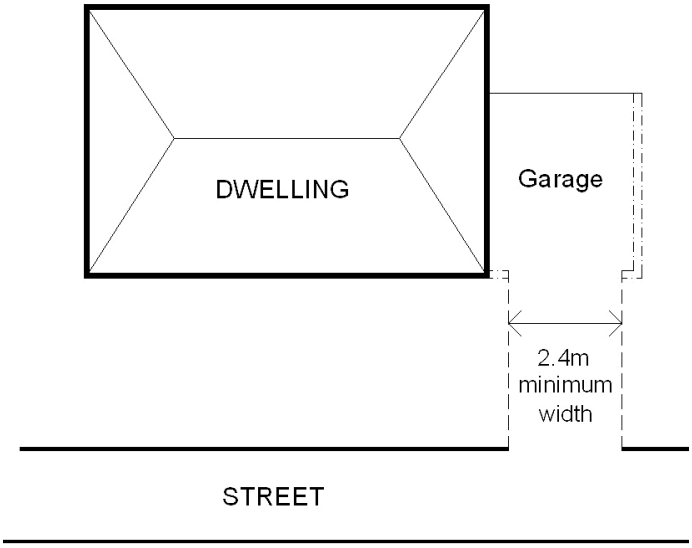
1. The dimensions of on-site car parking spaces ~~shall~~ must be in accordance with Australian Standard – AS/NZS 2890.1 (as amended) and Australian Standard – AS 2890.6.

2. All parking spaces must ~~shall~~ be designed to comply with the dimensional and manoeuvring requirements of the 85th percentile vehicle as defined by AS/NZS 2890.1 (as amended).

3. Parking spaces must~~shall~~ have a minimum clearance of 2.2 m from the finished floor level of a parking space and adjacent driveway area to any structure over a parking space.

4. Parking spaces adjoining walls and other structures or within single garages must~~shall~~ be 5.5 m long and 3.0 m wide with a clear garage opening of 2.75m. The garage opening (doorway width) may be reduced to 2.4m wide where the driveway leads straight into the garage (as shown in Figure 1). A double garage in a residential development shall be 5.5 metres long and 5.7m wide with a clear garage opening of at least 5m.

Figure 1



5. Parking spaces ~~shall~~ must have a grade no greater than 1:20.

4.3 Additional Controls for Development with Parking Ramps

1. This clause applies to all development where ramps are required for vehicle access to connect between substantially different levels.

2. Parking areas ~~shall~~ must be designed in accordance with *Australian Standard – AS/NZS 2890.1* (as amended), except where otherwise specified in this chapter.

3. Ramp grades within parking areas ~~shall~~ must be designed in accordance with the following table:

Ramp Length	Maximum Ramp Grade
Less than 20m	1:5 (20%)
Greater than 20m	1:6 (16.6%)

4. A 1.4m minimum transition length is required when a change in driveway and ramp grade equals 11% (1 in 9). Maximum changes of grade ~~shall~~ must not exceed 11% except where transitions comply with AS/NZS 2890.1.

4.4 Additional Controls for Adaptable Housing and Development Requiring Disabled Parking

1. The dimensions of onsite car parking for adaptable housing ~~shall~~ must be in accordance with *Australian Standard – AS 4299 Adaptable Housing* (as amended).

2. The minimum internal width for a double garage for an Adaptable Unit ~~shall~~ must be 6.2 metres.

3. As a minimum, parking for people with a disability ~~shall~~ must be provided, in accordance with the Building Code of Australia and the design and dimensions of such spaces shall be in accordance with the relevant Australian Standard.

4. Disabled parking spaces ~~shall~~ must have a grade no greater than 1:40, as detailed in AS/NZS 2890.1.

5. Where the Right of Carriageway/access driveway leads to an adaptable dwelling, then the minimum width of the paved driveway required ~~shall~~ must be 3.5m to accommodate an emergency vehicle as defined by the small rigid vehicle (SRV) in *Australian Standard AS/NZS 2890.2* (as amended). A turning head ~~shall~~ must be provided adjacent to the adaptable house to enable a small rigid vehicle to manoeuvre within the Right of Carriageway/access driveway and to proceed in a forward direction following a maximum three point turn. The intent of this control is to apply to adaptable dwellings served by an internal driveway/road and not dwellings that are located directly off a street, where emergency vehicles can park on the street.

6. Where a proposed driveway or circulation/access aisle in a basement carpark provides pedestrian access to an adaptable dwelling the driveway ~~shall~~ must be widened to the extent required to provide an accessible path of travel compliant with AS1428.1. This widened portion of the driveway or circulation/ access aisle shall be treated with a different colour or texture as required by *Australian Standard AS4299*. Alternatively, disabled car parking should be provided adjacent to or as close as possible to an accessible lift within a basement and the path of travel treated with a different colour or texture as required by AS4299.

4.5 Additional Controls for Tandem, Car Stackers, Car Lifts and Turntables

1. A queuing analysis report must be provided with parking and access details where development proposes mechanical parking installation such as Tandem, Car Stackers, Car Lifts and Turntables.
2. Installations of mechanical parking must be in accordance with AS5124:2017.
3. Maximum Tandem parking arrangement must be limited to a maximum of two spaces and maximum two for Car Stacker in the same level.
4. The maximum parking limit for spaces must not exceeded and these spaces must not be used for service vehicle parking, visitor or car share schemes.
5. The manoeuvring of and queuing for stacked vehicles must be able to occur wholly within the premises, without obstructing the entry or egress of other vehicles.
6. The minimum length of a tandem space must be 10.8 metre.
7. Car lifts will only be considered where it can be demonstrated that a basement ramp is not feasible/achievable.
8. Queuing analysis must be completed in accordance with Australian Standard AS2890.1 and Ausroads “Guide to Traffic Management”. The assessment is to establish whether the mechanical lift requires queuing to be accommodated on the property noting that queuing of vehicles on public roads will not be supported.

4.6 Design Requirements – Stacker & Car lift

1. The car stackers must be designed, constructed and managed in accordance with the following minimum standards.
    - Accommodate a B85 car, and
    - With a vehicle envelope height of 1.9m, and
    - A vehicle mass of 3000kg, and
    - Each parking plate must be independent access, and
    - Car park plate clear dimensions of 5.4m by 2.4m, and
    - Mechanical vertical lift / stabilizer arms / columns of the car stacker must not encroach within the minimum design parking envelope detailed in figure 5.2 of AS2890.1:2004.
2. The car lift must be designed in accordance with the following minimum standards.
    - Accommodate a B99 car, and
    - With a vehicle envelope height of 2.2m, and
    - A vehicle mass of 3500kg, and
    - Car park plate of 6m by 3m.
3. Provide traffic lights to control orderly access to the car lift. Line mark and sign temporary standing zones for motorist queuing to access the car lift.
4. Driveway access or ramps must have minimum width of 5.5m for two-way entry & exit. If it difficult to comply with clause 3.2.2 of AS2890.1 and access is controlled, boom gates, traffic lights, lifts then the design must comply with clause 3.5 of AS2890.1.
5. Turntable radius must be 4m for a single car and 6m for two cars. The minimum clearance all sides for minimum B85 vehicle is to be 300mm and the turntable area must be flat surface.
6. All multi-unit residential car parking spaces must be 'EV-Ready'. An 'EV-Ready' car space requires the provision of a backbone cable tray and a dedicated spare circuit within an EV Distribution Board enabling future installation of a smart EV charger and cabling to the EV Distribution Board.

4.7 Design Requirements – Car with Caravan

1. The dimensions of "design vehicle" standard for a car with caravan must be modelled with respect to the maximum caravan width of 2.5m.
2. The manoeuvring must be shown in design with swept path diagrams to the Council's satisfaction. The exit from the parking space must be in forward direction. Adequate signage to be provided as required.

5. Vehicular Access and Driveways

Vehicular access across footpaths to roadways presents a potential point of conflict between vehicles, pedestrians and cyclists. As such, the design and location of vehicle footpath crossings requires careful consideration to ensure public safety is optimised.

Driveways provide vehicle access within the development site, and are designed to ensure pedestrian and driver safety.

Footpaths, by definition, are not designed for vehicle use, with the exception of shared bicycle and pedestrian footpaths which may be required, as appropriate. Footpath pavements are located within the footpath reserve of the road reserve. Footpaths need to be controlled and designed in a way so as to ensure that there is a safe access for pedestrians, and that pedestrian and vehicle conflicts are reduced.

5.1 Objectives

1. Maximise safety for residents and visitors to a development.
2. Minimise traffic congestion and ensure adequate traffic management system.
~~1.~~

~~2.~~3. Ensure efficiency of vehicle circulation. (such as adequate width of driveway/ramp and suitable sight distance)

~~3.~~4. Ensure the safe and orderly movement of traffic, pedestrians and cycling movement.

~~4.~~5. Minimise the impact of noise and glare from vehicle movements on dwellings within or external to the site

5. Restrict vehicular access to development sites from arterial roads to optimise traffic flows and public safety, and reduce conflict between vehicles, pedestrians and cyclists.

5.2 Vehicular Access Controls for All Development except Subdivision

1. Where development has two (2) or more road frontages, vehicular access ~~shall should~~must be from the lowest order road shown on Sutherland Shire Council's Road Hierarchy Map.

2. Where a site has more than one road frontages, an applicant may make a submission to Council to gain access from the higher order classified road shown on Sutherland Shire Council's Road Hierarchy Map. The submission ~~shall~~ must demonstrate that the proposed access provides safe and convenient access; supports the defined road hierarchy; that the proposed access location fits well within the surrounding streetscape; and the proposed access location will not have any negative consequences for adjoining landowners.

3. Where a development is on the lower side of the roadway or where basement car parking is proposed, the driveway must~~is to~~ be a maximum grade of 5% for 3 m immediately inside the boundary to provide suitable sight lines and stopping distance for pedestrian safety.

4. Developments ~~should~~ must minimise potential conflicts between pedestrians and vehicles in the design and use of driveways, roadways and footpaths, and by separating pedestrian and vehicles movements.

5.3 Vehicular Access Controls for Subdivision Creating New Public Roads

1. New roads ~~shall~~ must comply with the following widths:

Public Roads	Minimum Reserve Width (m)	Minimum Carriageway width (m)	Footpath Reserve Width (m)	Max no. dwellings served
Minor Cul-de-sac (<6 sites)	12.5	5.5	3.5	20
Minor local street	14.5	7.5	3.5	75
Local Street	16.0	9.0	3.5	150
Collector and distributor	18.0	11.0	3.5	>150
Perimeter	20m	9.0	Variable	N/A

2. Bush fire prone lands that have Cul-de-sac are to have a minimum 12.5m radius to access fire trucks considering Heavy Rigid Vehicle (HRV).
3. The design for access of fire trucks in bush fire prone lands are to be carried out in accordance with AS2890.2, NSW Fire Safety Guideline – Access for Fire Brigade vehicles, NSW Rural Fire Service and Council standards.

~~2.~~4. Identified or planned bus routes and roads acting as perimeter roads in bush fire interface areas must~~are to~~ have a minimum 9 metre carriageway.

~~3.~~5. Proposed roads ~~shall~~ must comply with the following gradients:

Public Roads	Longitudinal Gradient Max (%) [1]	Grade Min (%)	Max Desirable Vehicular Speed (km/hr)	Stopping sight distance speed (km/hr)
Minor Cul-de-sac sites (<6 sites)	16	1.0	15-25	30
Minor Local Street	16	1.0	20-30	40
Local Street	16	1.0	20-30	40
Collector and Distributor	16	1.0	30-40	60
Perimeter	16	1.0	40-60	60

1 - Maximum gradient applies to the critical (inner) wheel path on the inside of the curve.

~~4.~~6. The gradient on a minor cul-de-sac which may serve up to and including 5 dwellings may extend up to 18% where it can be demonstrated that provides benefits in terms of minimising excessive earthworks while ensuring appropriate vehicle and pedestrian access and access for Council's waste disposal vehicle, NSW Fire Brigade and NSW Rural Fire Service vehicles. Special

provision shall be required in such circumstances for pedestrian access along the road, including stairs and/or stairs and handrails.

5.4 Vehicular Access Controls for Subdivision that Creates Internal Lots with Access Handles and/or Rights of Carriageway

1. Access handles to internal allotments ~~are to~~must be treated to the same standard as Rights of Carriageway.

2. Internal allotments must~~are to~~ be able to provide adequate on-site parking areas sufficient to cater for three vehicles together with turning facilities within each lot. The parking area is to be separate from the Right of Carriageway.

3. The reserve width for a Right of Carriageway or an access handle includes any easement for services.

4. All Rights of Carriageway/ access handles must~~shall~~ be 3.0m in width, with concrete pavement with kerb and relief drainage within a 3.65m reserve width to serve a maximum of three dwellings.

5. Proposed Rights of Carriageway/access handle serving 3 dwellings or less must~~shall~~ comply with the following longitudinal gradient maximum of 25%, gradient minimum of 1%, maximum desirable speed 0-5 km/hr and stopping sight distance speed of 20 km/hr. The maximum gradient applies to the critical (inner) wheel path.

6. Rights of carriageway must~~shall~~ comply with the requirements of a Circulation Roadway as described in AS/NZS 2890.1 (as amended).

7. In the re-subdivision of existing allotments, the potential for redevelopment on adjoining lands must~~is to~~ be considered. An overall subdivision plan may be prepared by Council if necessary.

8. A Right of Carriageway must ~~shall be permitted to~~ service no more than ~~only~~ three allotments, in addition to the road frontage allotment.

9. Where three (3) internal allotments are proposed, the minimum width of the allotments burdened by the Right of Carriageway shall be 22m including the Right of Carriageway. The Right of Carriageway shall be 4m wide including a 3m carriageway and a 1m service strip, except where the NSW Fire Service or NSW Rural Fire Service requirements exist.

10. Access to the front allotment ~~is to~~must be via the Right of Carriageway, unless otherwise specified by Council.

11. The design of the Right of Carriageway ~~shall ensure~~must ensure that all vehicles can exit in a forward direction at all times.

12. The Right of Carriageway for internal allotments ~~is to~~must comply with AS/NZS 2890.1 (B85 design vehicle). Where the length exceeds 50m or contains sight line restrictions the turning area shall accommodate a small rigid vehicle as defined by Australian Standard AS/NZS 2890.2. Where access to the site for NSW Rural Fire Service or NSW Fire Brigade vehicles is required, passing bays and turning heads shall comply with the requirements of those services. Where any waste disposal vehicle is to gain access to the Right of Carriageway the turning head shall be designed to accommodate Council's standard waste service vehicle.

13. Where a proposed Right of Carriageway gradient is more than 12.5% and less than 20%, and no separate pedestrian facility exists, provision for steps and a handrail ~~shall~~ must be made.

14. Where a proposed Right of Carriageway gradient exceeds 20% a separate pedestrian access stair with handrails shall be provided. Paths and steps ~~shall~~ must be a minimum 1.0 metre wide.

15. The minimum vertical clearance above the finished surface level of a Right of Carriageway or any associated turning head ~~shall~~ must be 4.5m.

5.5 Driveway Design Controls for All Developments

1. The dimensions of on-site driveways giving access to parking spaces ~~shall~~ must be in accordance with Australian Standard – AS/NZS 2890.1 (as amended), except where otherwise provided by this chapter. A driveway is classified as a circulation roadway as described by AS/NZS 2890.1 (as amended).

5.6 Additional Driveway Design Controls - Where Waste Service Vehicle Access to Site is proposed

1. Where an internal driveway is a through driveway with entry and exit on separate streets and waste service vehicles are to enter the site, all manoeuvring and passing bays must be designed to accommodate the swept manoeuvring area of Council's standard waste service vehicle.

5.7 Pedestrian Access Controls for All Development within the Site

1. Where a proposed pathway/footpath pavement gradient is more than 12.5% and less than 20%, provision for steps and/or handrail ~~shall~~ must be made. Where a proposed pathway/footpath pavement gradient equals or exceeds 20% a separate pedestrian access stair with handrails shall be provided.

2. Paths and steps ~~shall~~ must be a minimum 1.0 m wide within development sites.

5.8 Additional Pedestrian Access Controls for All Development within the Site except Dwelling Houses

1. The design of footpaths within car parking areas ~~shall~~ must be in accordance with *Australian Standard – AS/NZS 2890.*1 (as amended) except where otherwise provided by this chapter.

2. The use of contrasting pavement material ~~shall~~ must be used to define pedestrian access and vehicular areas.

3. The pedestrian access path of travel ~~shall~~ must be compliant with AS2890.6/ AS1428.1 and shall be separate from all other activities, and ~~shall~~ must allow uninterrupted travel by people with a disability.

5.9 Public Domain Pedestrian Access Controls for All Development except Dwelling Houses

1. Paved pedestrian footpaths ~~shall~~ must be provided within road reserves. Such footpaths shall have a minimum width of 1.2m.

2. Where footpath pavement is required in accordance with the Sutherland Shire Bicycle Network Map, then such must ~~shall~~ be provided in accordance with the Public Domain Design Manual.

3. Proposed pathways/footpaths must ~~shall~~ have a gradient of 12.5% or less. The maximum gradient applies to the inner curve radius.

4. A formal footpath width may be reduced or not required where a non-residential use (e.g., bushland) abuts the development site and services are not required along that footpath.

Note:

Until the publication of AS2890.6 refer to Australian Standard 2890.1-1993 for percentage of parking spaces required for parking for people with a disability.

6. Loading and Unloading Facilities

6.1 Objectives for All non-residential Development Requiring Loading and Unloading Facilities

1. Ensure appropriately designed loading and unloading facilities that are easily accessible to delivery vehicles and provide efficient manoeuvrability for vehicles.

6.2 Controls for All Non-Residential and Shop Top Housing Development Requiring Loading and Unloading Facilities

1. This clause applies to all development for the purposes of business or commercial premises, retail development, industrial, warehouse or bulky goods developments, hotels and motels, community facilities, medical facilities and places of public worship and other uses provided for within the ~~RTA Guide to Traffic Generating Development~~ TfNSW Guide to Transport Impact Assessment or all other development requiring a dedicated loading and unloading facility.

2. As a minimum standard, developments ~~shall~~ must be designed in accordance with the loading and service vehicle requirements within AS/NZS 2890.2 and the ~~RTA Guide to Traffic Generating Developments~~TfNSW Guide to Transport Impact Assessment, and ~~shall~~ must~~should~~ be suitable for a heavy rigid vehicle as defined by AS/NZS 2890.2.

3. All loading, unloading and manoeuvring of vehicles muat ~~shall~~ take place within the curtilage of the site, and vehicles are to enter and exit the site in a forward direction at all times.

4. Adequate separation of service vehicles from other vehicular and pedestrian traffic must ~~shall~~ be required, including the provision of defined pedestrian pathways in parking areas, and where loading and service vehicle movements occur.

5. Loading areas must ~~shall~~ be located so as to:
   a. reduce on-street loading;
   b. be freely available for use at all times.

6. Access to loading areas must ~~shall~~ be via rear laneways, where possible. Access and manoeuvring from the laneways shall comply with AS/NZS 2890.2 (as amended).

7. Where it is demonstrated that a vehicle smaller than a heavy rigid vehicle can adequately service the development, a restrictive covenant under Section 88 of the Conveyancing Act must ~~shall~~ restrict future uses to those that can be serviced by that class of vehicle.

8. Council may require a Loading Dock Management Plan for non-residential or shop top housing developments.

~~7.~~

6.3 Design Requirements – Industrial/Warehouse

1. Each industrial factory/unit must be provided with a loading bay.
2. Provision must be made for all loading and unloading to take place wholly within the designated loading area.

3. No loading or unloading must be carried out across parking spaces, landscaped areas, pedestrian aisles or on roadways.
4. Where it is proposed to service the site with articulated vehicles exceeding 12.5m in length, swept turning paths must be provided for that vehicle type.
5. The maximum longitudinal grade and cross fall must be 5%

Table: A

Industrial factory/units	**Required loading Dock size (AS2890.2)**
0m2 - 399m2	Small Rigid Vehicle (SRV)
400m2 – 1499m2	Medium Rigid Vehicle (MRV)
1500m2+	Heavy Rigid Vehicle (HRV)

6.4 Design Requirements – Multi-dwelling/Residential Flat Building

Each development must be provided with a loading bay as listed below:

a) Provision shall be made for all loading and unloading to take place wholly within the development.
b) No loading or unloading shall be carried out across parking spaces, landscaped areas, pedestrian accessways.

Table: B

Numbers of Units	**Required loading Dock size (AS2890.2)**
1-4	Small/Medium trucks
5-10	Small Rigid Vehicle (SRV)
10+	Medium Rigid Vehicle (MRV)

c) Where a shared loading bay is proposed to be within the driveway/aisle, the driveway/aisle must have minimum 6m to enable vehicle passing.
d) For developments greater than 20 Units a separate dedicated loading bay must be provided.
e) The maximum longitudinal grade and cross fall is to be 5%.

7. Provision of Facilities for Cycling

7.1 Objectives

1. To reduce car dependency and to support public transport use, development will give priority to walking, cycling and public transport access.

2. To promote and encourage the use of bicycles as a sustainable, safe and convenient means of transport for trip purposes that include shopping, education (to school), work, health and recreation.

3. To provide safe and secure cycle parking facilities that encourage the use of bicycles for the undertaking of a wide range of trips that include shopping, travelling to school and commuting to work.

4. To provide safe and secure cycle parking facilities in locations that reinforce its preference as an alternative to private motor vehicle use.

5. To take account of the Sutherland Shire Bicycle Network Map when determining the location of facilities for cycling and the connections between development sites.

Note:
The Sutherland Shire Bicycle Network Map is a strategy document which forms part of this DCP. The Network map shows where sites are located in relation to the network of active transport routes across the Sutherland Shire. More detail is available on Council's website on Shiremaps and in the Public Domain Design Manual.

7.2 Controls for All Development, Except Dwelling Houses, Dual Occupancies, and Multi Dwelling Housing

1. As a minimum, developments must provide 1 bicycle parking space per 10 car parking spaces for first 200 car spaces, then 1 space per 20 parking spaces thereafter. In addition, 1 unisex shower is required per 10 employees.

2. Bicycle parking facilities ~~are to~~must be installed in accordance with Australian Standard AS2890.3 – Bicycle Parking Facilities (as amended), Austroad's Guide to Traffic Engineering Practice – Part 14 – Bicycles and the Austroads Bicycle Parking Facilities: Guidelines for Design and Installation (AP-R527-16).

3. Bicycle parking facilities must address the following design principles:
    a. Accommodate all usual types of bicycles such that damage to them is minimised during storage and retrieval.
    b. Not pose a hazard to bicycle users, pedestrians or motorists.
    c. Be well lit, safe and secure, easy to access and use.
    d. Cater for the different needs of residents, employees and visitors to the development.

e. Be located in convenient and accessible locations within the development that allow for good passive surveillance; such as near key building entrances, the lobby and the lift core.
f. When located within a car park, ~~preferably~~ be situated at street level and in a manner that provides the most direct, safe and convenient access while minimising conflict with vehicles and pedestrians.
g. Where a bicycle parking and storage facility cannot be located at street level, it must be located no more than one level above or below street level. Access to street level entry and exits must be direct, safe and minimise potential conflicts with vehicles.

4. Development and works in the public domain must ~~shall~~ have regard to the Sutherland Shire Bicycle Network Map and the requirements of the Public Domain Design Manual.